One repository, every lab's cells.

Free for labs that share, a membership for teams that use the data, and a place where cell data converts.

← all documents · Kat Titterton · San Francisco

Vision

Every lab that images cells should know where its cells sit among everyone else's, and every dataset made from cells should carry that answer with it. Datum Line is one shared repository of raw cell images from every lab, on one map, and the product is a membership to it. Labs join free by sharing the control images they already take, and keep a share of what their data earns. Teams that use the data pay: screening teams first, for a check on every batch they run; AI teams the most, for aligned images from every lab by API. The arrow after that is left open on purpose.

The habit comes first, the way Strava's segments place a run among everyone who ran the same stretch of road. Here the road is the cell line, and the field is every lab that ran it. Everything below serves one rule: **uploading must be effortless, and the answer must be worth looking at.**

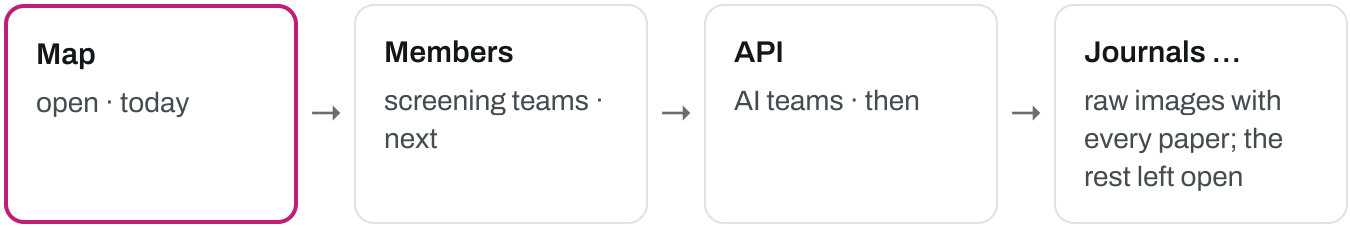
01 Why now

Cell data has a price. One vendor pays its experts more than $1.5M a day, about $85 an hour on average, at a $10B valuation. [10] Roche and Genentech paid Recursion a second $30M milestone for an accepted whole-genome map of microglial immune cells; $213M from the partnership to date. [11] A $1.8B virtual-cell data initiative, part new money and part existing funding; Meta, Google DeepMind and Isomorphic put in $300M together and get a year of exclusive access first. [12]

Cell data does not convert. EU-OPENSCREEN ran one Cell Painting protocol at 4 sites, with cells, serum and dye lots supplied centrally. Within a site, 84 to 94% of compounds replicate; across sites, 54 to 68% pair, and the authors call the profiles "not directly comparable between imaging sites" [13]. On the public JUMP data, a classifier names the lab from DMSO wells alone at 99.9% against 9.1% chance. Whoever makes cell data comparable sets its exchange rate.

02 The path

Fig. 1 **A starting point, not an end state**



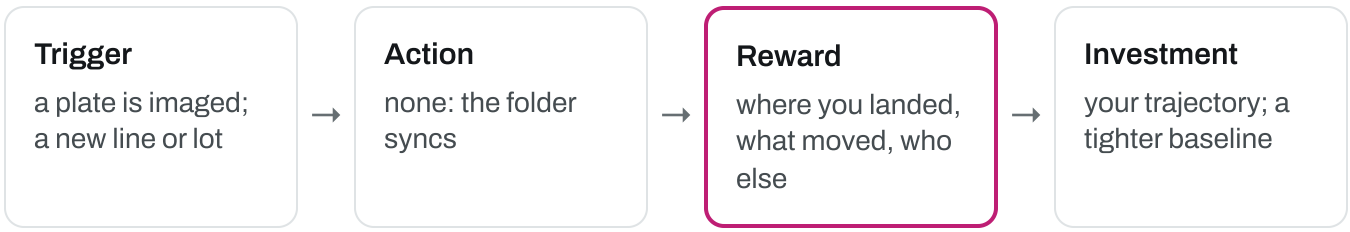
each stage pays for the next, and none is promised a date

STAGE	WHAT IT IS	WHO PAYS	WHAT HAS TO BE TRUE	STATUS
Map	every lab's DMSO wells placed on one map; the box, the code and the public reference	nobody: open, and never crippled	labs upload the controls they already image	today: 11 labs, 129 batches, 1,871 plates of public data
Members	the repository, and a check on every batch: where the cells sit, which lab they look like, what moved	teams that screen all year, $5–15k per cell line per quarter; companies that buy cell data, $10–50k per dataset	the map holds for the line	next: the first paying member, a screening team; prices proposed on board A15
API	aligned raw images from every lab, each traced to the lab that made it	AI labs and model builders, by use, and the most; each lab gets a share	labs choose to share; the map makes labs agree	then

STAGE	WHAT IT IS	WHO PAYS	WHAT HAS TO BE TRUE	STATUS
…	what the map makes possible next: journals asking for raw control images with every paper, alerts when a lot drifts across labs, a graded benchmark, models trained on every lab's controls			open

03 The habit loop

Fig. 2 **Every upload makes the next answer worth opening**



↺ back to Trigger: the reference grows, the next reading is sharper

STEP	IN DATUM LINE	DESIGN RULE
Trigger	the microscope finishes a plate; a new line or lot arrives	the trigger is the instrument, not an email
Action	nothing: the export folder syncs	any click is a leak; the reference starves on friction
Reward	where you landed, which features moved, who else moved	real uncertainty: nobody knows their position until they look
Investment	your history becomes a trajectory; your baseline tightens	leaving costs your own past, which is honest switching cost

04 Six magic moments

Each moment turns an image the lab already has into something it could not see before, and each is ordered by when a new member meets it.

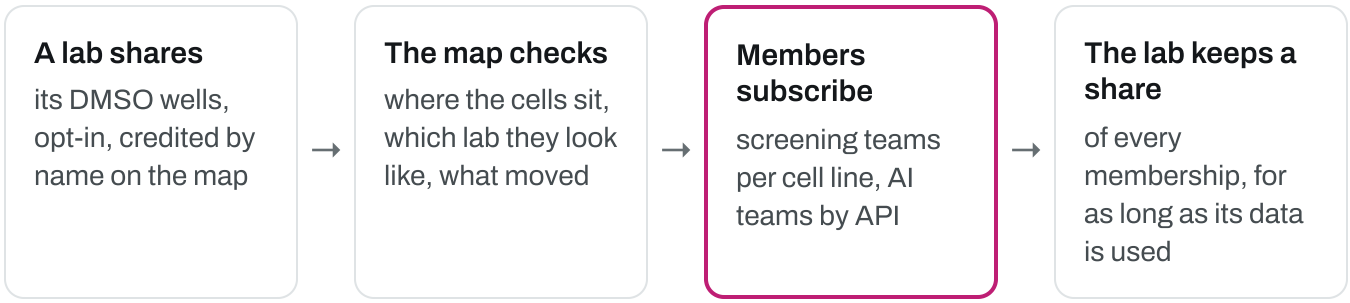
#	MOMENT	WHAT THE LAB SEES	WHEN
1	**Your past, diagnosed**	old plates dragged in on day one appear as a path across the field map: "you left the envelope in March"	first ten minutes
2	**Show the cells, not the score**	when a plate moves, its nuclei next to their nearest neighbours from labs in distribution: same size, more texture	every flag
3	**The fix**	what the in-distribution labs do differently: seeding density, media or serum lot	after a flag
4	**A forecast**	"at your current drift, this line leaves the envelope in about five weeks"	after three readings
5	**Someone else's plate saves your screen**	a lot you use is drifting at other labs, before your own plates show it	once labs share lots
6	**The check as currency**	a verifiable link for a methods section, a CRO handoff, a grant report	when results leave the lab

Moments 1 to 4 work for a single lab on day one, against the public reference. Moments 5 and 6 exist only once the network does, and no competitor can copy them without the labs.

05 Membership

Membership is access to the repository for a period, not ownership of the data. A lab that shares joins free; everyone else pays by how they use it. The check on every batch is what screening teams pay for first; aligned images from every lab are what AI teams pay the most for. A biotech’s first step costs nothing: a backup of its raw images, off-site and immutable, the copy cyber insurers ask for, encrypted with its own key; its control wells join the map when it chooses.

Fig. 3 **Every membership pays the labs that made the data**



↺ every lab that shares widens the map, and a wider map makes every check sharper

WHO JOINS	WHAT MEMBERSHIP GIVES	PRICE, PROPOSED	ROLE
labs and core facilities	the repository and the map; a check on every batch; the leaderboard	free: sharing controls earns it	the map
biotech companies	their raw images backed up off-site and immutable, private to them; their controls join the map when they choose	free	the hook
teams that run cell screens all year	a check on every batch they run, against every lab on the same line	$5–15k per cell line per quarter	first revenue
AI labs and model builders	aligned raw images from every lab, by API	the most: priced by use, not yet set	main revenue
companies that buy cell data	each dataset they buy, placed on the map	$10–50k per dataset	data revenue
benchmark partners	a test set with faults planted on purpose	$50–200k per year	the standard

WHAT MEMBERS GET	WHY IT IS WORTH PAYING FOR
The check	the comparison a lab cannot make alone: on the public data, 6 of 129 batches sat off the map, in 3 of 11 labs (results/e9)

WHAT MEMBERS GET	WHY IT IS WORTH PAYING FOR
Aligned data	it converts: on the map, agreement between labs on the same drugs goes from 0.75 to 0.78 (board A14)
Early access	a window before data goes public, which consortia already sell: a year, at the virtual-cell initiative [12]
The benchmark	AI labs and model builders graded on planted faults, with a cell-count baseline on every task (board A16)

What never gets sold. A lab's treated data, its results, its name without consent, a ranking of instruments, or a verdict for hire. Membership buys access and the check; it never buys a better verdict. Prices and shares are proposals (boards A15 and A22) until the first partner labs and the first members set them.

06 Trust, without the token

The open-science half of the product, and the reason a lab would share.

PROMISE	HOW	STATUS
Control wells only	nothing else is shared: no compound, no target, no result; the frame is applied on the lab's side	today
A private backup	raw images a company backs up with us are encrypted with its own key; Datum Line cannot read them	proposed
Every lab leaves a fingerprint	99.9% of DMSO wells name their own lab on plates the model never saw, so a dataset claimed from one lab and made in another is caught	today
Signed and timestamped	every check signed, its fingerprint written to a public log: anyone can verify it, nobody can backdate it; a blockchain is one such log, not a requirement	proposed
The lab keeps its data	sharing is a choice and can be withdrawn; a share of every membership goes to the lab that made the data	proposed

PROMISE	HOW	STATUS
Known members	members verified at sign-up; unusual access flagged; every access in a log the members can read	proposed
Open code and data	the code is MIT on GitHub, the public data is JUMP's under CC0, and every number comes from a script anyone can rerun	today
Low risk by design	images of cell lines: no human genetic data and no pathogen data; the map holds DMSO wells only	today

For a decentralized-science audience this is the point: open data, verifiable records and contributor upside, without a token to speculate on and without betting on anyone's results.

07 The game

Every batch a lab sends is ranked against every batch the field has seen, on its DMSO wells alone: how typical the cells are, how alike the plates are, how steady the lab is from batch to batch, and cells per well against the field. Nothing in the score touches a treated well or a result, so the only way to climb is better culture and imaging practice. On the public data, lab 5 holds the field's longest streak, 33 batches (results/e14). A lab sees its rank, its personal best, its streak and the lab to beat.

DO	DON'T	WHY
rank practice, never results	rank a lab's results or its drugs	the only way up is better culture and imaging
labs are numbers until they choose a name	publish a lab's name or rank without consent	shame makes labs hide their worst plates, which teach the field most
credit for uploading flagged plates	penalise a batch for sitting off the map	a failure uploaded is worth more than a success
explain every flag with cells and a likely fix	show a bare red dot	a verdict without a reason reads as judgement
let the instrument trigger uploads	send nudges to protect a streak	the reward is the answer, not the visit

08 Metrics

North star: the share of a lab's plates that upload with no human action. If it is high, the loop turns; if people have to remember, nothing else matters.

METRIC	WHAT IT TELLS YOU
automatic upload share	the habit exists without effort
flagged plates uploaded, not deleted	labs trust the product with bad news
labs per line, lines per lab	the map is widening
labs sharing to the repository	labs trust the repository with their data
paying members, and how many renew	membership is worth paying for, again
share of each membership paid to the lab	contributors are earning, so more will share
checks cited outside the lab	the check has become currency

09 What could go wrong

RISK	WHY IT IS REAL	ANSWER
cold start	no map without labs, no labs without a map	seed from JUMP's public controls; U2OS first, any other line from its first batch
labs will not share	data is a lab's edge, and pharma data is intellectual property	a lab that keeps its images private still gets the check, as a paying member, the way screening teams do; sharing is free, optional and pays the lab
AI teams train on public data for free	JUMP and other public sets are open to anyone	the open map stays free; members pay for what public data lacks: every lab aligned, new batches as labs share them, the lab behind every image

RISK	WHY IT IS REAL	ANSWER
public data must stay free	NIH-funded labs must share the data behind their findings	charge members for the check, alignment and compute, never for access to data a lab owes the public
buyers want designed data at scale	the deals above buy factories and consortia, not leftovers	place factory and consortium data on the map too; the benchmark grades their models
a platform builds its own map	phenomics companies hold the most images	no one can check their own drift against themselves; the reference has to be neutral and open

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